Penn Engineering

EAS5450 Quiz
8:35 – 8:45am
Canvas Quizzes

"Dell business model turns to muscle as rivals struggle."

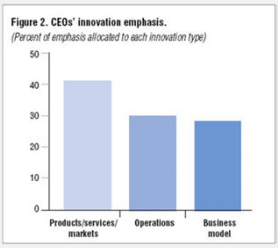
"It's a simple business model, but what makes it a success is doing it on that kind of scale and with that kind of complexity."

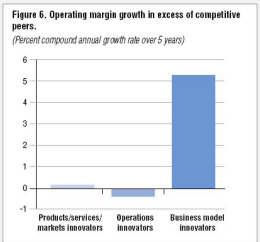
USA TODAY: 1/19,23/03

EAS5450 Class #11

1

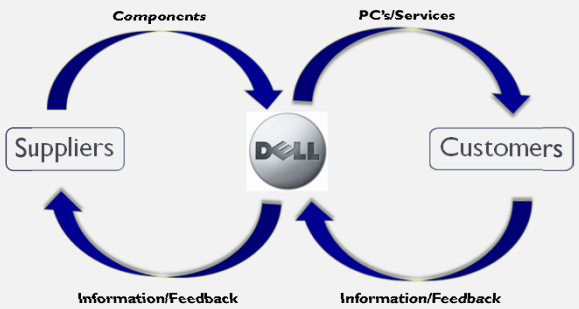
Business Model Innovation Critical to Competitive Advantage





IBM Global CEO Study, 2006.

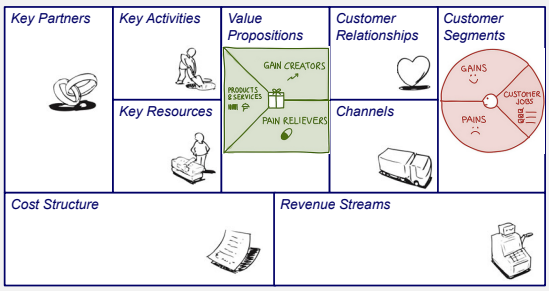
2



3

Recall:

The Business Model Canvas…



4

3 Successful Strategies

- ★ **Overall Cost Leadership**
- ★ **Differentiation**
- ★ **Focus**

Dell combines all 3!

5

Key Financial Ratios

XYZ COMPANY INCOME STATEMENT
(Dollars in thousands except per-share amounts)

		Year Ended June 30, 2007	Year Ended June 30, 2006
1.	Net Sales	$765,000	$725,000
2.	Cost of Sales	535,000	517,000
GROSS MARGIN		230,000	208,000
3.	Selling expenses	76,500	84,000
4.	General and administrative expenses	20,300	25,500
5.	Depreciation and amortization	28,060	25,000
OPERATING INCOME		105,140	73,500
6.	Dividend and interest income	5,200	10,000
7.	Interest expense	16,250	16,750
INCOME BEFORE TAXES		94,090	66,750
8.	Income taxes	41,400	26,250
9.	Extraordinary income (loss)	0	0
NET INCOME		$52,690	$40,500

XYZ COMPANY BALANCE SHEET
(Dollars in thousands)

		June 30, 2007	June 30, 2006
ASSETS:			
12.	Cash and cash equivalents	$18,900	$19,000
13.	Marketable securities	34,000	32,000
14.	Accounts receivable	165,000	150,000
15.	Less: allowances for doubtful accounts	(8,000)	(5,000)
16.	Inventories	160,000	165,000
17.	Prepaid expenses and other current assets	32,000	30,000
TOTAL CURRENT ASSETS		402,900	391,000
18.	Property, plant and equipment	263,000	246,000
19.	Less: accumulated depreciation	(123,000)	(97,000)
NET PROPERTY, PLANT & EQUIPMENT		140,000	149,000
20.	Goodwill and intangibles (goodwill, patents, etc.)	3,000	2,000
21.	Less: accumulated amortization	(200)	(200)
NET INTANGIBLES		2,800	2,010
22.	Other assets	0	0
TOTAL ASSETS		$545,700	$542,610
LIABILITIES AND SHAREHOLDERS' EQUITY			
23.	Accounts payable	$72,000	$65,000
24.	Notes payable	51,000	41,000
25.	Other accrued expenses	30,000	36,000
26.	Income taxes payable	15,000	16,000
27.	Current portion of long-term debt	8,000	
TOTAL CURRENT LIABILITIES		176,000	158,000
28.	Long-term debt	156,000	156,000
29.	Deferred income taxes	18,000	18,000
TOTAL LIABILITIES		$350,000	$332,000
30.	Preferred stock (7.5% cum.vative) $100 par value	9,000	9,000
31.	Common stock: $9 par value (20,000,000 shares authorized) 16,000,000 shares issued and outstanding 2007 16,500,000 shares issued and outstanding 2006	75,000	72,000
32.	Additional paid-in capital	20,000	15,000
33.	Retained earnings	89,700	114,610
TOTAL SHAREHOLDERS' EQUITY		$193,700	$210,610
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY		$543,700	$542,610

6

- **INVENTORY TURNOVER:**

$$\frac{\text{COGS}}{\text{Inventory}} \quad \text{or} \quad \frac{365}{\text{COGS/Inventory}}$$

XYZ Company 2007

$$\frac{535}{180} = 3X \text{ or } 123 \text{ Days}$$

vs 131 Days in 2006

- **COLLECTION PERIOD:**

$$\frac{\text{Accts Receivable}}{\text{Sales/365}}$$

$$\frac{162}{765/365} = 77 \text{ Days}$$

vs 76 Days in 2006

- **DAYS OF PAYABLES:**

$$\frac{\text{Accts Payable}}{\text{COGS/365}}$$

$$\frac{72}{535/365} = 49 \text{ Days}$$

vs 49 Days in 2006

7

- **CURRENT RATIO:**

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

XYZ Company 2007

$$\frac{406}{176} = 2.3\ X$$

vs 2.1X in 2006

- **QUICK RATIO ("ACID TEST"):**

$$\frac{\text{(Current Assets) - (Inventory)}}{\text{Current Liabilities}}$$

$$\frac{406 - 180}{176} = 1.3\ X$$

vs 1.1X in 2006

- **RETURN ON EQUITY (ROE):**

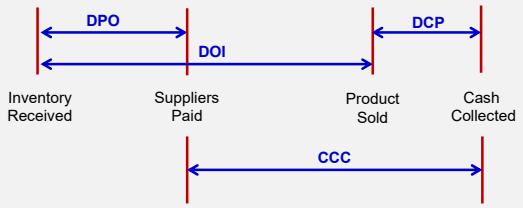
$$\frac{\text{Earnings}}{\text{Shareholders' Equity}}$$

$$\frac{52.7}{346} = 15.2\%\ p.a.$$

vs 13.3% in 2006

8

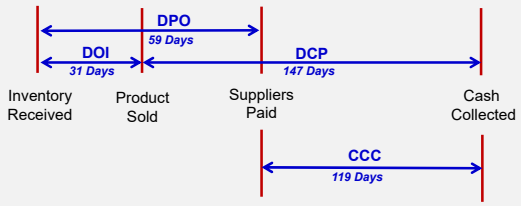
Cash Conversion Cycle



- DPO: Days Payables Outstanding = Accts Payable / (COGS/365)
- DOI: Days of Inventory = Inventory / (COGS/365)
- DCP: Days Collection Period = Accts Receivable / (Sales/365)
- **CCC = DOI + DCP - DPO**

9

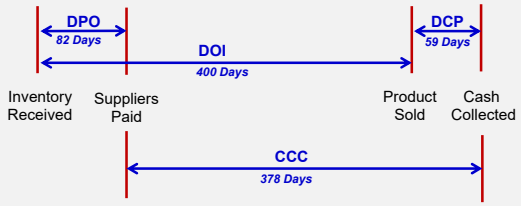
Cash Conversion Cycle



- DPO: Days Payables Outstanding
- DOI: Days of Inventory
- DCP: Days Collection Period
- **CCC = DOI + DCP - DPO**

10

Cash Conversion Cycle



- DPO: Days Payables Outstanding
- DOI: Days of Inventory
- DCP: Days Collection Period
- **CCC = DOI + DCP - DPO**

11

Key Financial Ratios

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-K

(Mark One)

☑ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended January 31, 2020

or

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from _____ to _____

Commission File Number: 001-37867

Dell Technologies Inc.

(Exact name of registrant as specified in its charter)

Delaware (State or other jurisdiction of incorporation or organization)

80-0890963 (I.R.S. Employer Identification No.)

One Dell Way, Round Rock, Texas 78682 (Address of principal executive offices) (Zip Code)

1-800-289-3355 (Registrant's telephone number, including area code)

12

DELL TECHNOLOGIES INC.
CONSOLIDATED STATEMENTS OF INCOME (LOSS)
(in millions, except per share amounts)

	Fiscal Year Ended		
	January 31, 2020	February 1, 2019	February 2, 2018
Net revenue:			
Products	$ 69,918	$ 70,707	$ 60,898
Services	22,236	19,914	18,142
Total net revenue	92,154	90,621	79,040
Cost of net revenue:			
Products	54,525	57,889	51,433
Services	8,606	7,679	7,070
Total cost of net revenue	63,221	65,568	58,503
Gross margin	28,933	25,053	20,537
Operating expenses:			
Selling, general, and administrative	21,319	20,640	18,560
Research and development	4,992	4,604	4,384
Total operating expenses	26,311	25,244	22,953
Operating income (loss)	2,622	(191)	(2,416)
Interest and other, net	(2,626)	(2,170)	(2,353)
Loss before income taxes	(4)	(2,361)	(4,769)
Income tax benefit	(5,533)	(180)	(1,843)
Net income (loss)	5,529	(2,181)	(2,926)
Less: Net income (loss) attributable to non-controlling interests	913	129	(77)
Net income (loss) attributable to Dell Technologies Inc.	$ 4,616	$ (2,310)	$ (2,849)

13

DELL TECHNOLOGIES INC.
CONSOLIDATED STATEMENTS OF FINANCIAL POSITION
(in millions)

	January 31, 2020	February 1, 2019
ASSETS		
Current assets:		
Cash and cash equivalents	$ 9,302	$ 9,676
Accounts receivable, net	12,484	12,371
Short-term financing receivables, net	4,895	4,398
Inventories, net	3,281	3,649
Other current assets	6,906	6,044
Total current assets	36,868	36,138
Property, plant, and equipment, net	6,055	5,259
Long-term investments	864	1,005
Long-term financing receivables, net	4,848	4,224
Goodwill	41,601	40,089
Intangible assets, net	18,107	22,270
Other non-current assets	10,428	2,835
Total assets	$ 118,861	$ 111,820
LIABILITIES, REDEEMABLE SHARES, AND STOCKHOLDERS' EQUITY (DEFICIT)		
Current liabilities:		
Short-term debt	$ 7,737	$ 4,320
Accounts payable	20,065	19,213
Accrued and other	9,773	8,495
Short-term deferred revenue	14,881	12,944
Total current liabilities	52,456	44,972
Long-term debt	44,319	49,201
Long-term deferred revenue	12,919	11,066
Other non-current liabilities	5,383	6,327
Total liabilities	115,077	111,566
Commitments and contingencies (Note 10)		
Redeemable shares (Note 17)	629	1,196
Stockholders' equity (deficit):		
Common stock and capital in excess of $0.01 par value (Note 14)	16,091	16,114
Treasury stock at cost	(65)	(63)
Accumulated deficit	(16,891)	(21,349)
Accumulated other comprehensive loss	(709)	(467)
Total Dell Technologies Inc. stockholders' deficit	(1,574)	(5,765)
Non-controlling interests	4,729	4,823
Total stockholders' equity (deficit)	3,155	(942)
Total liabilities, redeemable shares, and stockholders' equity (deficit)	$ 118,861	$ 111,820

Assets Dell Owns

Liabilities Dell Owes

Dell's Net Worth

14

- **INVENTORY TURNOVER:**

Dell Technologies 2020

$$\frac{\text{COGS}}{\text{Inventory}} \text{ or } \frac{365}{\text{COGS/Inventory}}$$

$$\frac{63.2}{3.3} = 19X \text{ or } 19 \text{ Days}$$

- **COLLECTION PERIOD:**

$$\frac{\text{Accts Receivable}}{\text{Sales/365}}$$

$$\frac{12.5}{92.2/365} = 49 \text{ Days}$$

- **DAYS OF PAYABLES:**

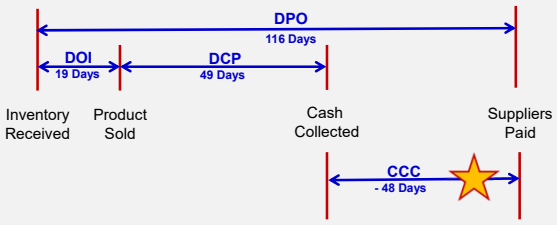
$$\frac{\text{Accts Payable}}{\text{COGS/365}}$$

$$\frac{20.1}{63.2/365} = 116 \text{ Days}$$

15

Cash Conversion Cycle

Dell Technologies 2020



16

Product Development at Dell

17

Transforming Product Development

OLD PROCESS <1993:

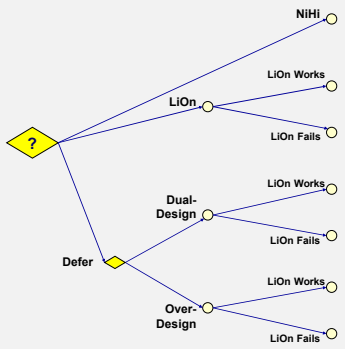
- Informal
- Autonomous Teams
- Unpredictable
- No Manufacturing Involvement
  - "Thrown over the wall"
- No Cross-Functionality

NEW PROCESS 1993+:

- Structured, Well-Defined
- Multi-Functional Product Teams
  - Electrical Engineers
  - Mechanical Engineer
  - Industrial Designer
  - Software Engineer
  - Product Marketer
- Formal "Stage-Gate" Phase Reviews
  - Profile Phase: define product & market
  - Planning Phase: business issues
  - Implementation Phase: alpha prototypes
  - Qualification Phase: beta tests
  - Launch Phase: customer buying experience
  - Acceptance Phase: 3 mos customer feedback
- Reduces Unpredictability
- Customer-Driven & Differentiated

18

Dell's Battery Decision



19

Dell's Decision Criterion…

Net Sales Revenues
- Cost of Goods Sold
= Gross Margin
- Development Costs
= "Net Margin"

Expected Value (EV) = $\sum_{i=1}^{n} (Probability_i)(Value_i)$

20

NiHi Decision

11,000,000 Units/Yr x 2.5% Dell Share x 3 Yrs = 825,000 Units

$2,400 Avg Price x 25% Gross Margin = $600 Gross Margin/Unit

825,000 Units x $600 GM/Unit = $495,000,000 Gross Margin
- 10,000,000 Development Cost
= $485,000,000 Net Margin

x 100% Probability

$485,000,000 Expected Net Margin

21

LiOn Decision – If It Works

11,000,000 Units/Yr x **3.0%** Dell Share x 3 Yrs = **990,000** Units

$2,400 Avg Price x 25% Gross Margin = $600 Gross Margin/Unit

990,000 Units x $600 GM/Unit = $594,000,000 Gross Margin
- 10,000,000 Development Cost
= $584,000,000 Net Margin

x **60%** Probability

$350,400,000 Expected Net Margin

22

LiOn Decision – If It Doesn't Work

11,000,000 Units/Yr x 2.5% Dell Share x **50%** x 3 Yrs = 412,500 Units

$2,400 Avg Price x 25% Gross Margin = $600 Gross Margin/Unit

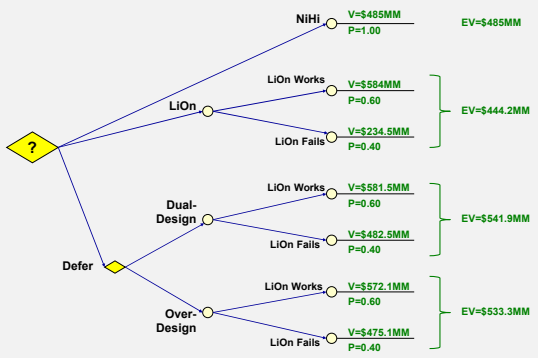
412,500 Units x $600 GM/Unit = $247,500,000 Gross Margin
- 10,000,000 Development Cost
- **3,000,000 Re-Work**
= $234,500,000 Net Margin

x **40%** Probability

$93,800,000 Expected Net Margin

23

Dell's Battery Decision



24

Epilog:

Hey dude, your Dell is on fire!

Man blames Dell laptop for house fire

engadget
June 22, 2006

25

Dell Battery Recall

Dell Recalls 4.1 Million Batteries Due to Fire Hazard

TechNewsWorld - 15 Aug 2006

Dell Laptop Battery Recall May Cost $400 Million

Bloomberg - 15 Aug 2006

Japan Orders Sony, Dell To Investigate Battery Problems

Redmond Channel Partner - 24 Aug 2006

Virgin Atlantic has become the third airline to restrict the use of Apple and Dell laptop batteries on its flights.

Reg Hardware - 14 September 2006

26

THE WALL STREET JOURNAL.

BUSINESS | Updated February 6, 2013, 8:26 p.m. ET

Boeing Preps Redesign to Get 787 Flying Again

By JON OSTROWER, ANDY PASZTOR and DANIEL MICHAELS

Associated Press

Boeing 787 jets are lined up this week at an airfield in Washington state.

Boeing Co. is proposing a series of battery design changes that it believes would minimize the risks of fire on its 787 Dreamliners and allow the grounded jets to fly again while it continues searching for a longer-term fix, say government and industry officials briefed on the matter.

The company is looking at increasing the separation between cells in the lithium-ion batteries to reduce the potential hazards from heat or fire spreading within the batteries and adding enhanced heat-sensors, these officials said. Boeing also is considering ways to keep cells more rigid, preventing them from shifting under certain conditions and interfering with electronics.

The Explosive History of Lithium-Ion Batteries

Boeing grounded 50 of its 787s on Jan. 16 after a battery fire forced the emergency landing of an All Nippon Airways flight. The dangerous and costly failure of the lithium-ion batteries made plenty of headlines—and not for the first time. —*Shawn Hazzo*

Bloomberg Businessweek, Feb 3, 2013

27

TECHNOLOGY NEWS | Fri Sep 16, 2016 | 11:02am EDT

Samsung to replace or refund one million U.S. Galaxy Note 7 phones

By Ankur Banerjee

Samsung Electronics Co Ltd (005930.KS) formally recalled 1 million Galaxy Note 7 smartphones sold in the United States, replacing or refunding the flagship phones, whose susceptibility to catching fire has damaged the image of the Korean powerhouse.

Samsung received 92 reports of batteries overheating in the United States, including 26 reports of burns and 55 cases of property damage, the company said as it announced the recall in cooperation with the U.S. Consumer Product Safety Commission (CPSC).

The recall is a costly setback for Samsung, which was counting on Galaxy Note 7 to bolster sales as rivals such as Apple Inc (AAPL.O) launch new devices. The scale of the recall is unprecedented for Samsung, the world's largest smartphone maker.

28

THE WALL STREET JOURNAL.

THURSDAY, FEBRUARY 1, 2007 - VOL. CCXLIX NO. 26 ★★★ $1.00

- Dell known for its business model:
  - Supply chain strength
  - Direct sales
- Stalled as PC buying behavior changed.
- Missed sales and earnings projections.
- Rollins: stern, aloof, single-minded.
- Dell: hands-on, MBWA, empowers team, motivational leader, sense of urgency.
- Dell stock up $1/share (4.1%) on announcement.

Dell's Founder Returns as CEO As Rollins Quits

Computer Firm Expects To Miss Earnings Mark; Stock Rises After Hours

By Christopher Lawton And Joann S. Lublin

Facing a deepening strategic quandary at the personal-computer giant he founded 22 years ago, Michael Dell, 42, replaced his former protégé, Kevin Rollins, as chief executive of Dell Inc.

Mr. Dell's move to resume control of the company he founded in his dorm room as a start-up in his college dormitory into one of the world's largest PC makers, is a sharp reversal for Austin, Texas-based Dell, Texas, company—known for its business model of selling PCs directly to consumers and business clients over the phone and Internet—has stalled in recent years as the business it dominates the PC world has changed dramatically.

Mr. Dell's renewed day-to-day involvement may do little to solve the company's problems, at least in the short term. It comes amid a sharp shake-up, the company also disclosed Thursday that its profits and revenues for the fiscal fourth quarter ending this week to fall short of its earlier forecasts.

Dell's reversal has taken place largely since Mr. Dell handed the reins to Mr. Rollins in 2004. Mr. Dell remained chairman of the board, but, in an interview, Mr. Dell largely declined to comment on Mr. Rollins's departure, and Mr. Rollins couldn't be reached for comment.

Mr. Rollins, a longtime great friend and colleague for many years and made huge contributions for the company during a critical period, [but] I felt that it was a critical time for the company and I felt that I should step back

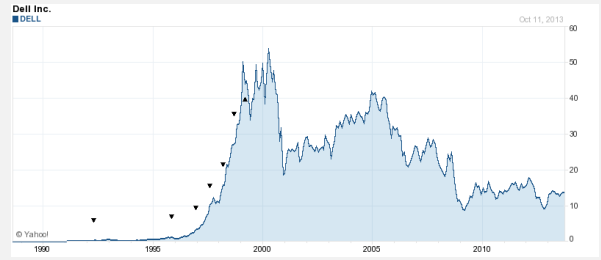
Please turn to page A2

Michael Dell

29

Dell Common Stock Performance

Splits: 4/10/92 [3:2], 10/30/95 [2:1], 12/9/96 [2:1], 7/28/97 [2:1], 3/9/98 [2:1], 9/8/98 [2:1], 3/8/99 [2:1]



IPO

Our Case
Battery Decision

Retail to Consumers
Revenues up; Earnings down; Quality problems

Targeting Corporations
$890M revenue; Portables launched

Retirement
Michael Dell steps down

Encore
Michael returns

Gone Private

30

Dell 5-Year Stock Performance

vs NASDAQ

31

Dell to Sell Itself for $24.4 Billion

By BEN WORTHEN, IAN SHERR and ISHIRA OVIDE

Dell Inc. on Tuesday struck a $24.4 billion deal to take itself private that it called a turning point in efforts to boost its value, even as some shareholders said they are unhappy with the deal's premium.

Founder and CEO Michael Dell—along with Silver Lake Partners—will offer Dell's holders a per-share price of $13.65 in cash. The stock closed Tuesday at $13.42 on Nasdaq.

In a securities filing, Dell didn't disclose details of its funding arrangements, including the contribution of Silver Lake or Mr. Dell's investment firm or the amount of debt the parties aim to raise.

Reuters

Dell founder and chief executive Michael Dell.

The deal "immediately delivers value to shareholders," Brian Gladden, Dell's chief financial officer, said on Tuesday in a brief interview. While Dell had made progress in turning itself around, "we do recognize that it will take more time" and is better achieved without the scrutiny and limitations of being a publicly traded company, Mr. Gladden said.

Sanford C. Bernstein analysts wrote that the "majority of shareholders appear likely to approve" the deal, citing an estimate that the top 25 holders, after Mr. Dell, control 42% of shares.

Less than two hours after the deal was announced, rival Hewlett-Packard Co. was quick to criticize the takeover, saying the buyout would add uncertainty for customers and limit Dell's ability to invest in new products.

Dell will continue to be based in Round Rock, Texas. The deal is expected to close in the second quarter of Dell's 2014 fiscal year.

Wall Street Journal: Feb 5, 2013

32

Dell is now a private company.

On September 12, 2013, Dell stockholders approved the proposal in which Michael Dell, Dell's Founder, Chairman and CEO, will acquire Dell in partnership with global technology investment firm Silver Lake Partners. The merger transaction closed on October 29, 2013, and the company has commenced the process to delist its common shares from the NASDAQ Stock Market. Per the merger agreement, Dell shareholders are entitled to receive $13.75 in cash, in addition to a special dividend of $0.13 per common share.

As a private company, we are going back to our roots, to the entrepreneurial spirit that made Dell one of the fastest growing, most successful companies in history. We're unleashing the creativity and confidence that have always been the hallmarks of our culture. We will be able to serve our customers with a single-minded purpose and drive the innovations that will change our world for the better.

33

COMMENTARY

Going Private Is Paying Off for Dell

A year later, we're able to focus on customers and the long term, rather than activist investors.

By Michael Dell
Nov. 24, 2014 6:47 p.m. ET

25 Comme

My partners at Silver Lake Management and I successfully took Dell private a year ago in the largest corporate privatization in history. I'd say we got it right. Privatization has unleashed the passion of our team members who have the freedom to focus first on innovating for customers in a way that was not always possible when striving to meet the quarterly demands of Wall Street.

To be clear, public markets can and do play an integral role. In Dell's formative years, public markets provided capital enabling us to grow and thrive, as is the case for many companies. The problem is when customer and shareholder interests diverge. At Dell, we faced a confluence of factors in making the decision to end a 25-year run as a publicly traded company. These factors included the big opportunities ahead, the required pace of innovation and investment, and an affliction of short-term thinking that drove a wedge between our customer and investor priorities.

Wall Street Journal, Nov 24, 2014

34

Dell Closes $60 Billion Merger with EMC

'We feel very well positioned both in the new areas of technology and in the existing areas of technology,' says CEO Michael Dell

Dell completed its acquisition of EMC Technologies on Wednesday. PHOTO: EUROPEAN PRESSPHOTO AGENCY

By Rachael King
Updated Sept. 7, 2016 10:15 a.m. ET

Dell Inc. Wednesday said it completed its $60 billion deal to acquire EMC Corp., the largest technology merger in history.

The new company, to be named Dell Technologies, aims to be a one-stop shop for information technology sold to business. The company employs about 140,000 people globally and will maintain operations in Hopkinton, Mass., where EMC was located. With $74 billion in revenue, Dell Technologies will be the world's largest privately controlled tech company.

Wall Street Journal: Sept 7, 2016

35

BARRON'S

TECHNOLOGY | REAL TIME ANALYSIS

Dell Technologies Will Go Public in December After Tracker Deal Approved

By Andrew Bary Dec. 11, 2018 11:08 am ET

Dell Technologies will go public later this month, after getting approval Tuesday to buy out shareholders of its tracking stock for VMware in a deal valued at about $21 billion.

Holders of more than 61% of the Dell Technologies class V tracking stock (ticker: DVMT) backed the controversial cash-and-stock deal that will result in a windfall of about $12 billion to Michael Dell and Silver Lake, the current owners of Dell, which was taken private in a 2013 leveraged buyout. The deal needed the approval of a majority of the holders of the 199 million tracking shares outstanding.

Under the terms of the transaction, Dell plans to pay about $14 billion in cash and issue about $7 billion of equity to the tracker investors at the closing expected on Dec. 28. The newly issued Dell class C common shares (DELL) are expected to start trading on the New York Stock Exchange on Dec. 28.

36

After 5 years as a private company,
Dell opened on the NYSE at $46/Shr on 12/28/2018
= $23/Shr post 11/1/2021 split

[4pm Closing Price 2/19/25 = $120.98/Shr]

Yahoo Finance 2/14/2025

37